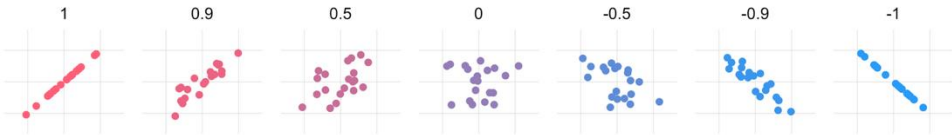


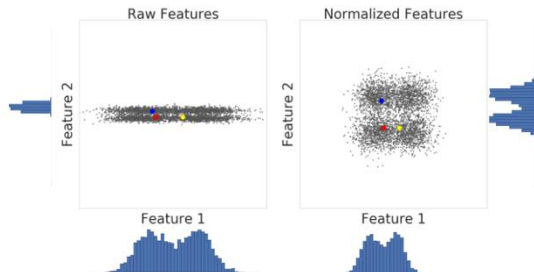
COMS20017 – Algorithms & Data



Majid Mirmehdi

Lecture MM-03

This lecture



- Data acquisition
- Data characteristics: distance measures
- **Data characteristics: summary statistics**
- **Data normalisation and outliers**

Mean and Variance

For one-dimensional data $\mathbf{x} = \{x_1, \dots, x_n\}$,

Mean: [average]

$$\mu = \frac{1}{N} \sum_i x_i$$

Variance: [spread]

$$\sigma^2 = \frac{1}{N-1} \sum_i (x_i - \mu)^2$$

Standard Deviation:

$$\sigma = \sqrt{\frac{1}{N-1} \sum_i (x_i - \mu)^2}$$

Mean and Covariance

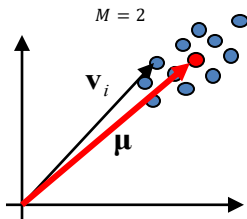
For multi-dimensional data:

e.g. M dimensions with $\{\mathbf{v}_1, \dots, \mathbf{v}_N\}$, i.e. there are N vectors/datapoints where each vector has M elements.

Mean vector:

Computed independently
for each dimension

$$\boldsymbol{\mu} = \frac{1}{N} \sum_i \mathbf{v}_i$$



Covariance:

Gives spread and how
variables change together

$$\mathbf{C} = \frac{1}{N-1} \sum_i (\mathbf{v}_i - \boldsymbol{\mu})^2$$

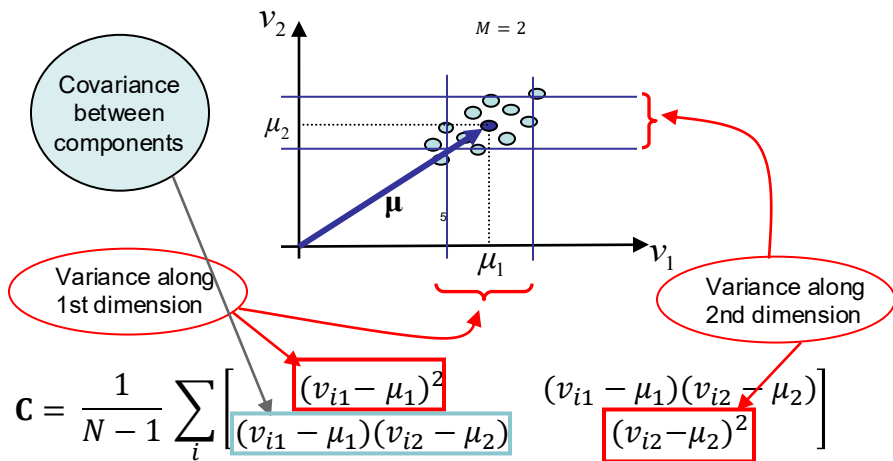
$$\mathbf{C} = \frac{1}{N-1} \sum_i (\mathbf{v}_i - \boldsymbol{\mu})^T (\mathbf{v}_i - \boldsymbol{\mu})$$

$$\mathbf{C} = \frac{1}{N-1} \sum_i \begin{bmatrix} (v_{i1} - \mu_1)^2 & (v_{i1} - \mu_1)(v_{i2} - \mu_2) \\ (v_{i1} - \mu_1)(v_{i2} - \mu_2) & (v_{i2} - \mu_2)^2 \end{bmatrix}$$

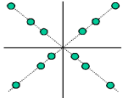
N when the population mean is known, $N-1$ when not!

Mean and Covariance

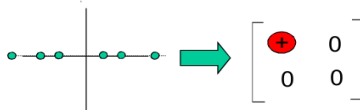
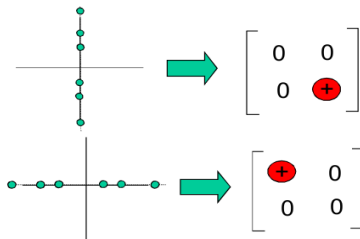
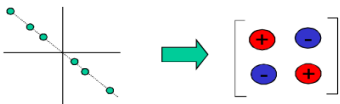
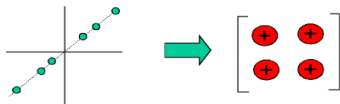
$$\boldsymbol{\mu} = \frac{1}{N} \sum_i \mathbf{v}_i$$



Covariance Matrix

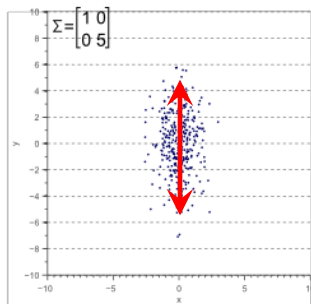
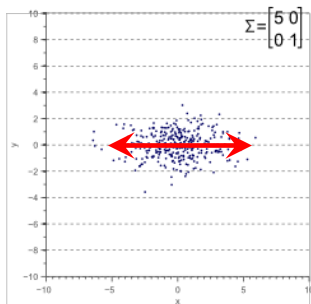
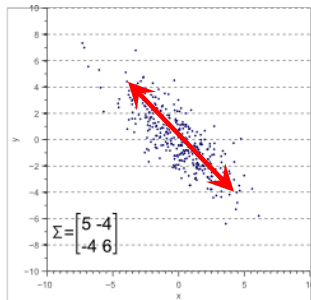
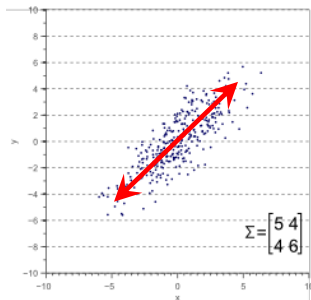
$$\mathbf{C} = \frac{1}{N} \sum_i \begin{bmatrix} (v_{i1} - \mu_1)^2 & (v_{i1} - \mu_1)(v_{i2} - \mu_2) \\ (v_{i1} - \mu_1)(v_{i2} - \mu_2) & (v_{i2} - \mu_2)^2 \end{bmatrix}$$



$$\begin{bmatrix} + & 0 \\ 0 & + \end{bmatrix}$$



Spread and Covariance

- The shape of the data is defined by the covariance matrix.
- Diagonal spread is captured by the covariance, while axis-aligned spread is captured by the variance.



Covariance Matrix

In three dimensions,

$$\mathbf{C} = \frac{1}{N} \sum_i \begin{bmatrix} (v_{i1} - \mu_1)^2 & (v_{i1} - \mu_1)(v_{i2} - \mu_2) & (v_{i1} - \mu_1)(v_{i3} - \mu_3) \\ (v_{i1} - \mu_1)(v_{i2} - \mu_2) & (v_{i2} - \mu_2)^2 & (v_{i2} - \mu_2)(v_{i3} - \mu_3) \\ (v_{i1} - \mu_1)(v_{i3} - \mu_3) & (v_{i2} - \mu_2)(v_{i3} - \mu_3) & (v_{i3} - \mu_3)^2 \end{bmatrix}$$

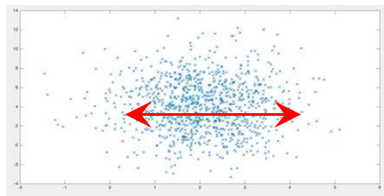
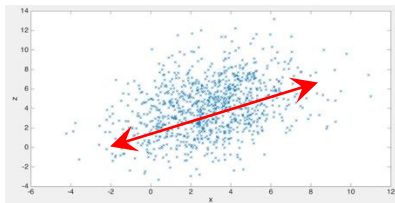
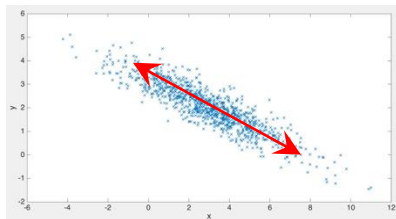
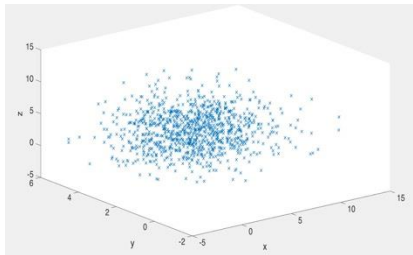
A Covariance matrix is always:

- ▶ square
- ▶ symmetric
- ▶ variances on the diagonal
- ▶ covariance between each pair of dimensions in non-diagonal elements

Covariance Matrix example

For the covariance matrix,

$$\mathbf{c} = \begin{bmatrix} 5 & -2 & 2 \\ -2 & 1 & 0 \\ 2 & 0 & 7 \end{bmatrix}$$



Correlation

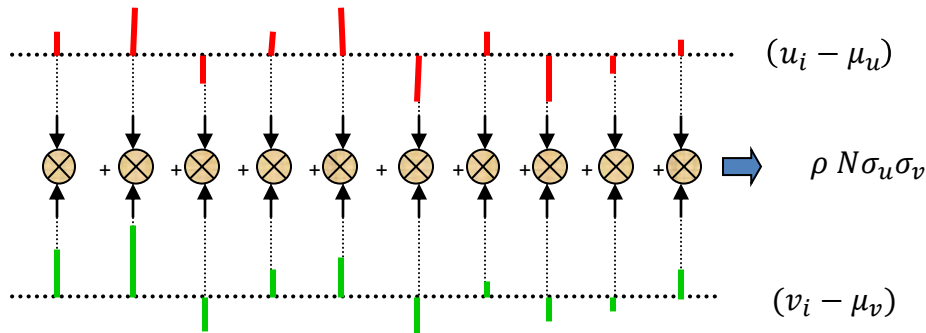
- We are often interested in the degree of similarity between two sequences in terms of their variation (independent of their absolute values)
- Correlation of two sequences u and v of length N

$$\rho = \frac{1}{N\sigma_u\sigma_v} \sum_{i=0}^{N-1} (u_i - \mu_u) (v_i - \mu_v)$$

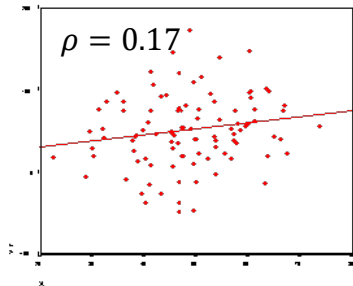
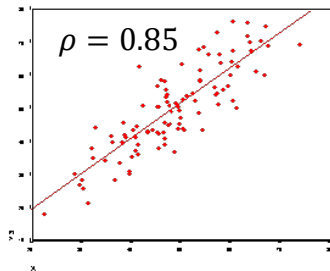
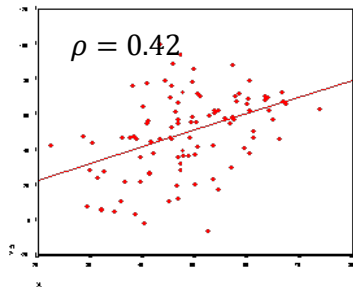
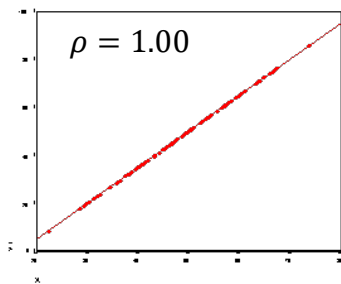
Division by variance product $\sigma_u\sigma_v$ normalises measure to be independent of absolute value and is unitless $\rightarrow$ captures similarity in variation (or *structure*).

Example: Correlating Coefficient for Two Sequences

$$\rho = \frac{1}{N\sigma_u\sigma_v} \sum_{i=0}^{N-1} (u_i - \mu_u) (v_i - \mu_v)$$



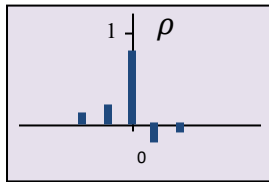
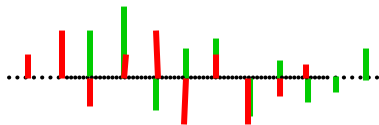
Correlation Example



Correlation – shifting similarity

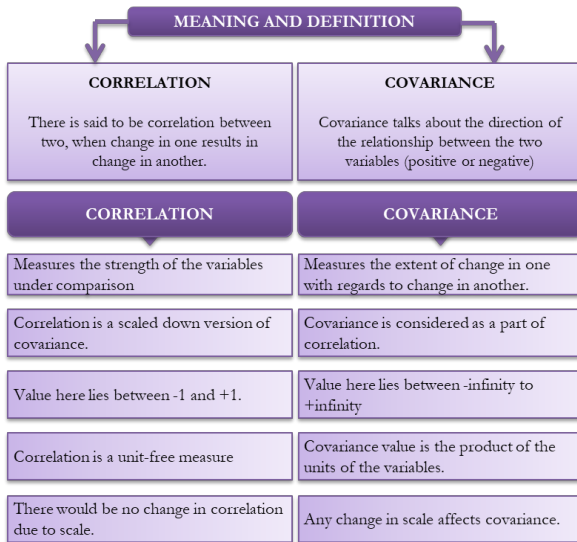
- Sometimes sequences are similar after applying a shift.
- Can be measured with **cross-correlation**

$$\rho = \frac{1}{N\sigma_u\sigma_v} \sum_{i=0}^{N-1} (u_i - \mu_u)(v_{i-j} - \mu_v)$$



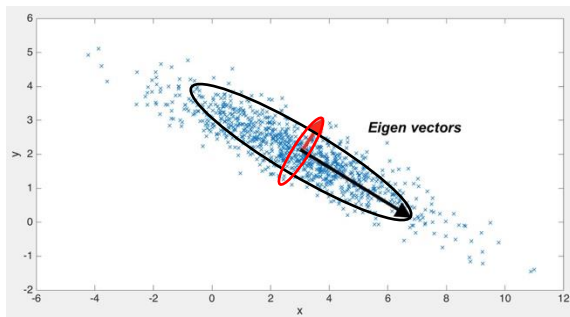
'most likely shift' given by position j of maximum value in cross-correlation

Self Study: Correlation and Covariance



Eigen analysis

- Eigenvectors and eigenvalues define the principal axes and spread of points along directions
- **Major axis** - eigenvector corresponding to larger eigenvalue (i.e. larger variance)
- **Minor axis** - eigenvector corresponding to smaller eigenvalue (i.e. smaller variance)
- These can be represented using major and minor axes of ellipses



Eigen analysis

Definition

For a square matrix $\mathbf{C}$,
if there exists a non-zero column vector $\mathbf{v}$ where

$$\mathbf{C}\mathbf{v} = \lambda\mathbf{v}$$

then,

$\mathbf{v} \rightarrow$ eigenvector of matrix $\mathbf{C}$

$\lambda \rightarrow$ eigenvalue of matrix $\mathbf{C}$

e.g.

$$\mathbf{C} = \begin{bmatrix} 0 & -1 \\ 2 & 3 \end{bmatrix}, \mathbf{v}_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \lambda_1 = 1$$

Eigen analysis

- To calculate eigenvectors of a square matrix, solve $\mathbf{C}v = \lambda v$

$$\longrightarrow |\mathbf{C} - \lambda \mathbf{I}| = 0$$

where

- $\mathbf{I}$ is the identity matrix
- $|\mathbf{C}|$ is the determinant of the matrix

For 2×2 matrices, there are two eigenvalues λ_1, λ_2

$$\mathbf{C} - \lambda \mathbf{I} = \begin{bmatrix} 0 & -1 \\ 2 & 3 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} = \begin{bmatrix} -\lambda & -1 \\ 2 & 3 - \lambda \end{bmatrix}$$

$$|\mathbf{C} - \lambda \mathbf{I}| = \lambda^2 - 3\lambda + 2 = (\lambda - 1)(\lambda - 2)$$

$$\lambda_1 = 1 \quad \text{and} \quad \lambda_2 = 2$$

Eigen analysis

- ▶ After the eigenvalues are found, the eigenvectors can be calculated

For $\lambda_1 = 1$

$$\begin{bmatrix} 0 & -1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} v_{11} \\ v_{12} \end{bmatrix} = \begin{bmatrix} v_{11} \\ v_{12} \end{bmatrix} \quad (2)$$

- ▶ This simplifies to:

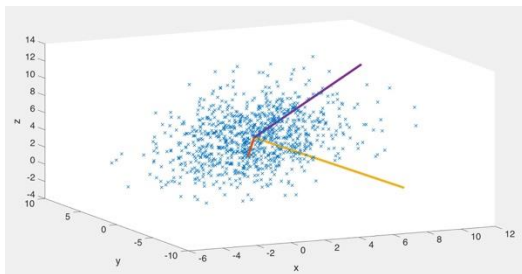
$$\begin{bmatrix} -v_{12} \\ 2v_{11} + 3v_{12} \end{bmatrix} = \begin{bmatrix} v_{11} \\ v_{12} \end{bmatrix} \quad (3)$$

- ▶ If we set $v_{12} = 1$, then we get the eigenvector:

$$\begin{bmatrix} v_{11} \\ v_{12} \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix} \quad (4)$$

- ▶ Verify that this is indeed a valid eigenvector by calculating $Cv = \lambda v$

3D example



➤ Eigenvalues $\rightarrow \quad \lambda_1 = 0.08 \quad \lambda_2 = 4.52 \quad \lambda_3 = 8.40$

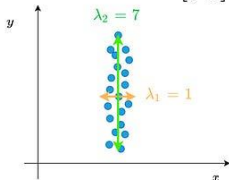
➤ Eigenvectors $\rightarrow \quad v_1 = \begin{bmatrix} -0.42 \\ -0.90 \\ 0.12 \end{bmatrix} \quad v_2 = \begin{bmatrix} 0.71 \\ -0.40 \\ -0.57 \end{bmatrix} \quad v_3 = \begin{bmatrix} 0.57 \\ -0.15 \\ 0.81 \end{bmatrix}$

➤ Principal/Major axis is v_3 (corresponding to the largest eigenvalue)

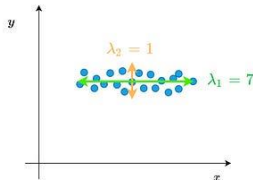
Eigenvalue and Variance

If the covariance matrix of the data is a diagonal matrix, i.e the covariances are zero, the variances are equal to the eigenvalues.

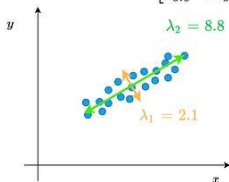
1 $C = \begin{bmatrix} 1 & 0 \\ 0 & 7 \end{bmatrix}$ $\lambda_{1,2} = [1 \ 7]$
 $V = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$



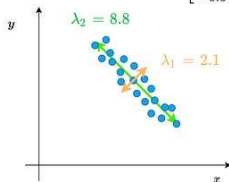
2 $C = \begin{bmatrix} 7 & 0 \\ 0 & 1 \end{bmatrix}$ $\lambda_{1,2} = [7 \ 1]$
 $V = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$



3 $C = \begin{bmatrix} 4 & 3 \\ 3 & 7 \end{bmatrix}$ $\lambda_{1,2} = [2.1 \ 8.8]$
 $V = \begin{bmatrix} -0.8 & -0.5 \\ 0.5 & -0.8 \end{bmatrix}$



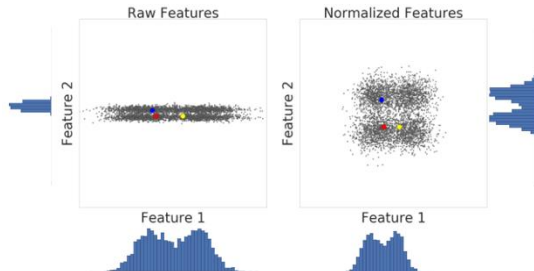
4 $C = \begin{bmatrix} 4 & -3 \\ -3 & 7 \end{bmatrix}$ $\lambda_{1,2} = [2.1 \ 8.8]$
 $V = \begin{bmatrix} -0.8 & 0.5 \\ -0.5 & -0.8 \end{bmatrix}$



λ = eigenvalues

V = eigenvectors

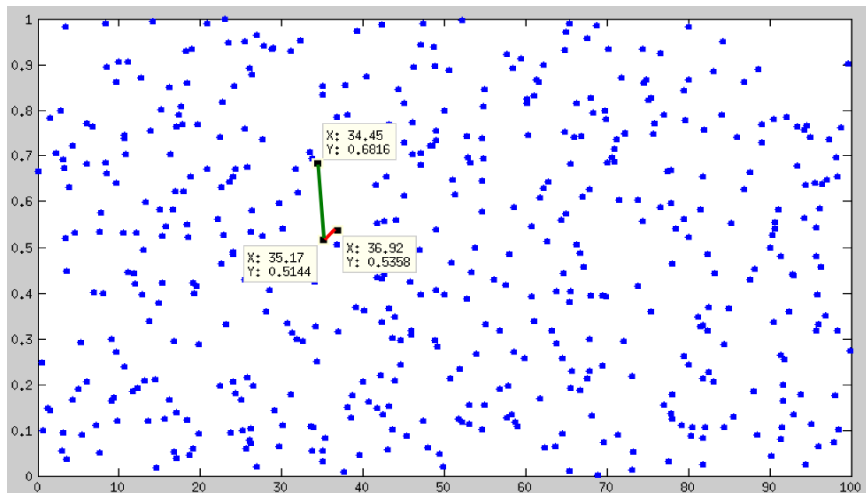
Next



- Data acquisition
- Data characteristics: distance measures
- Data characteristics: summary statistics
- **Data normalisation and outliers**

Data Characteristic - Data Normalisation

- Note the difference in magnitude between the two dimensions below!
- Data may need to be normalised before distance is calculated



Data Characteristic - Data Normalisation

► Methods for normalisation:

1. Rescaling
$$x' = \frac{x - \min(x)}{\max(x) - \min(x)}$$

rescales the range of features to [0, 1]

2. Standardisation (also known as z-score)

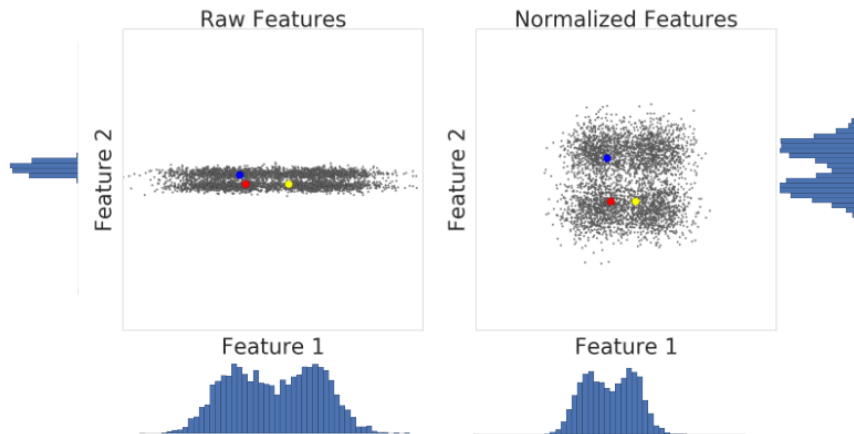
$$x' = \frac{x - \mu}{\sigma}$$

makes the values of each feature in the data have zero-mean and unit-variance

3. Scaling to unit length
$$x' = \frac{x}{\|\mathbf{x}\|}$$

scales components of feature vector so that the complete vector has length one

Normalisation Example



Brief return to Distance Measures

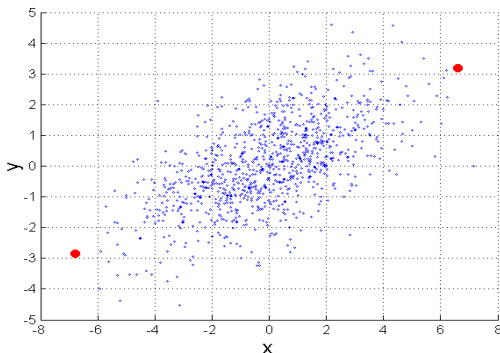
Mahalanobis Distance is a measure of distance between a data vector and a set of data, or a variation that measures the distance between two vectors from the same dataset:

$$\text{mahalanobis}(a, b) = \sqrt{(a - b)^T \Sigma^{-1} (a - b)}$$

$$\text{where } \text{cov}(X, Y) = \Sigma = \frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})$$

Warning: Σ is the covariance matrix of the input data D

For **red** points, the Euclidean distance is 14.7, and the Mahalanobis distance is 6.



Brief return to Distance Measures

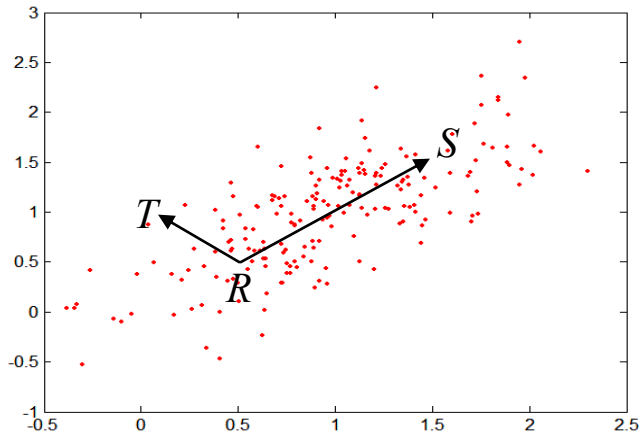
Mahalanobis Distance example:

Given $R = (0.5, 0.5)$, $S = (1.5, 1.5)$, $T = (0.0, 1.0)$, find the Mahalanobis distance RS and RT .

$$\Sigma = \begin{bmatrix} 0.3 & 0.2 \\ 0.2 & 0.3 \end{bmatrix}$$

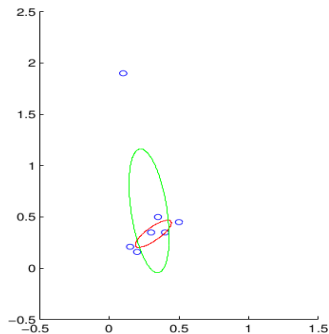
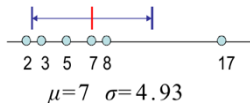
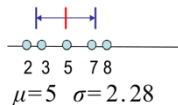
$$RS = 2$$

$$RT = \sqrt{5}$$



Data Characteristics - Outliers

- Mean, variance and covariance can provide concise description of 'average' and 'spread', but not when outliers are present in the data
- **outliers**: An *outlier* is an observation that lies an abnormal distance from other values in a random sample from a population.
- usually due to fault in measurement and not always easy to remove

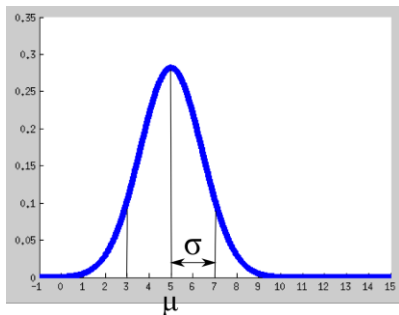


Normal or Gaussian Distribution (Reminder)

For a normal distribution $N(\mu, \sigma^2)$ in one dimension, the probability density function (pdf) can be calculated as:

$$p(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

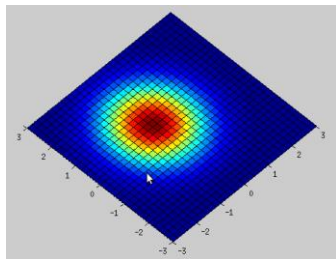
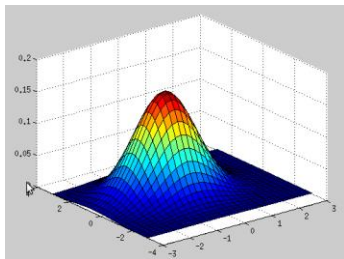
68% of data within 1σ of μ
92% within 2σ of μ
99% within 3σ of μ



Normal Distribution - Multi-dimensional (reminder)

For multi-dimensional normal distribution $N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, the probability density function (pdf) can be calculated as

$$p(\mathbf{x}) = \frac{1}{2\pi\|\boldsymbol{\Sigma}\|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right)$$



WARNING: Σ here is the capital letter of σ and represents the covariance matrix. It is NOT the summation sign!

Some of the topics next in COMS20017

- Least Squares and Regression
- Clustering data
- Classification of data
- The Fourier transform
- Principal Components Analysis
- Convolutions